Department of Chemical Engineering

Chemical Reaction Engineering II (CL24303)

MO2026

Tutorial 1

Date Assigned	14.08.2026	Submission Date and Time	23.08.2026 23:59

1. A list of some industrial catalytic processes has been shared on 04th August. By no means is it exhaustive, hence you are encouraged to find more. Thereafter, please find out: 3
    (i) Commercial name and shape of the catalysts
    (ii) Synthesis and characterisation techniques
    (iii) Type of reactors used
    (iv) Manufacturer (Companies) and their presence in India

2. Calculate the external surface area per unit volume of spherical Zeolite Y catalyst used in Fluidised Catalytic Cracker (FCC) of 100 microns in diameter. What size would these particles need to be if the external surface is to be 320 m^2/g? The density of the particles is 1.5 g/cm^3 [Ans: 3×10^4 m^2/m^3, 6.25×10^{-7} cm] 2

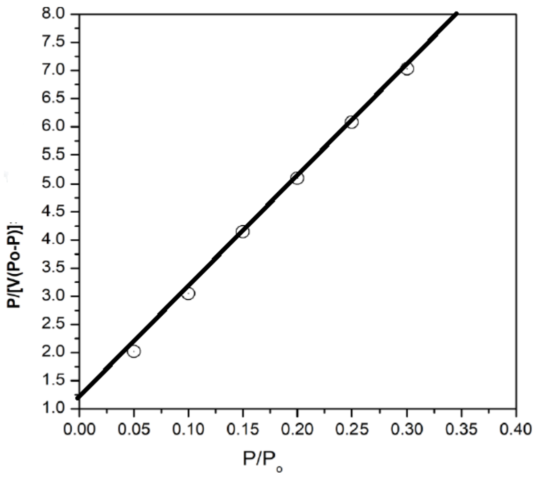
3. The following data on CO_2 adsorption loading on Zeolite-13x are reported at 40 and 100 °C (Sep. Purif. Technol. 85 (2012) 17–27).

T= 40 °C		T= 100 °C	
P (atm)	q (mmol/gm)	P (atm)	q (mmol/gm)
0.01	0.01	0.02	0.45
0.02	0.5	0.15	1.1
0.13	2.4	0.4	1.9
1.1	3.6	1.2	2.7
2.5	4	2.5	3.4
3.8	4.2		

Evaluate the maximum loading (q_{max}) and the adsorption equilibrium constant (K) using the equation of Langmuir adsorption isotherm:

$$q = \frac{q_{max}Kp}{1 + Kp}$$

4. From the Brunauer-Emmett-Teller plot shown below, estimate the surface area per gram of the silica gel. The quantity on y-axis has units 10^{-3} g/cc. Use the data for the adsorption of (a) nitrogen at -195.8°C (α_{N2}=16.2 × 10^{-16} cm^2) (b) Oxygen at -183 °C (α_{O2}=14.2 × 10^{-16} cm^2), and calculate the BET surface area (S_g) [Ans: (a) 209.65 m^2/g , (b) 204 m^2/g] 6



5. Example 8-4 of JM Smith 3

6. Example 8-5 of JM Smith 4